

Yi Gao

Robot Learning · Contact-Rich Manipulation · Humanoid Compliance

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Education

SWUFE-UD Data Science Institute

Sep 2024 – Jun 2028 (expected)

Southwestern University of Finance and Economics · Chengdu, China

Information Management and Information Systems; GPA: **87.7/100**

Four-year, China-based dual-degree program with the University of Delaware

University of Delaware degree: B.S. in Information Systems (expected)

Research Experience

SymBiosis

May 2026 – Present

Research Intern; ongoing remote collaboration · Advisor: Dr. Pengxiang Ding

SoftSONIC: Residual Learning for Humanoid Compliance

- Independently responsible for implementation, experimental design, and analysis. Developed a zero-initialized, 64-dimensional latent residual (~0.4M parameters) over a frozen SONIC tracker, using SoftMimic's compliant motion augmentation for PPO targets; no external-force input to the residual at inference.
- Investigated training failures with force-oracle ablations, open-loop residual fitting, closed-loop response probes, and reward-scale analysis; added configuration checks for termination thresholds and privileged-information separation.
- In preliminary standing-motion simulations using 1,024 parallel evaluation environments, increased the project-specific compliance-progress score from **0.322 to 0.683** and reduced the actual-to-desired force ratio from **3.262 to 1.912**, with adaptive sampling disabled.
- Observed similar compliance scores in two training seeds and with deterministic inference. Evaluation currently uses 10 standing clips with wrist perturbations; addressing zero-force tracking degradation and testing unseen-motion transfer remain ongoing work.

Lenovo Robotics Research Institute (Shanghai)

Jul 2025 – Dec 2025

Research Intern · Solution & Development Center

- Developed a context-grounded robot agent prototype for industrial patrol, integrating visual perception, location/time-specific safety rules, and action protocols into a traceable decision pipeline.
- Collected and organized **200+ real-world video clips** using a DEEPRobotics X30 quadruped; built a structured annotation and VLM evaluation workflow comparing QwenVL and GLM-family models.
- Drafted *CE4Patrol: Multi-Layer Context Reasoning for Industrial Anomaly Inspection* (manuscript; Yi Gao, et al.).

Selected Technical Project

ACT and Imitation Learning

Feb – Mar 2026

Built an ACT-style Transformer architecture with action chunking and reproduced an ACT training workflow in ManiSkill; implemented a VAE to study latent-variable learning.

Technical Skills

Programming & tools: Python, C++, PyTorch, Git, Linux, Docker

Simulation: Isaac Lab / Isaac Sim, MuJoCo, ManiSkill, PyBullet

Methods: PPO, residual policy learning, motion tracking, imitation learning, experimental diagnostics

Honors & Languages

Meritorious Winner, Mathematical Contest in Modeling (COMAP), 2026

Second Prize, National English Competition for College Students, 2026 · CET-4: 632; CET-6: 599